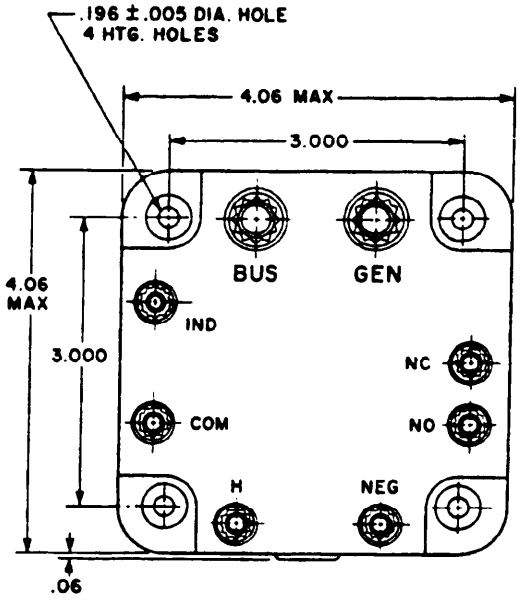
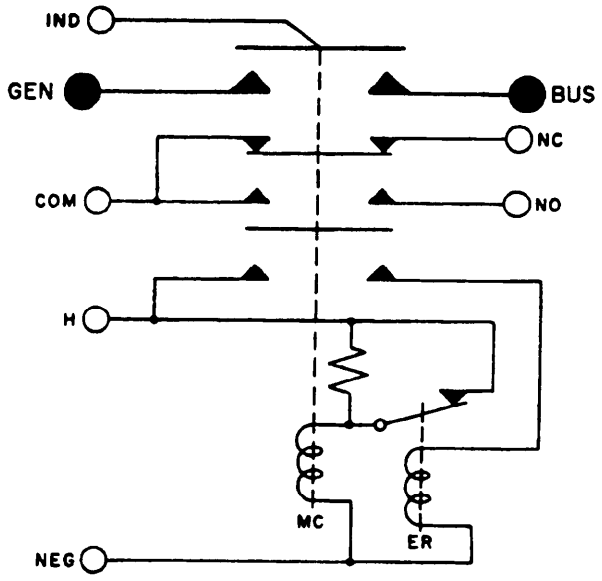
# MILITARY SPECIFICATION SHEET

## RELAY, ELECTROMAGNETIC, DIRECT CURRENT, 28 VOLTS

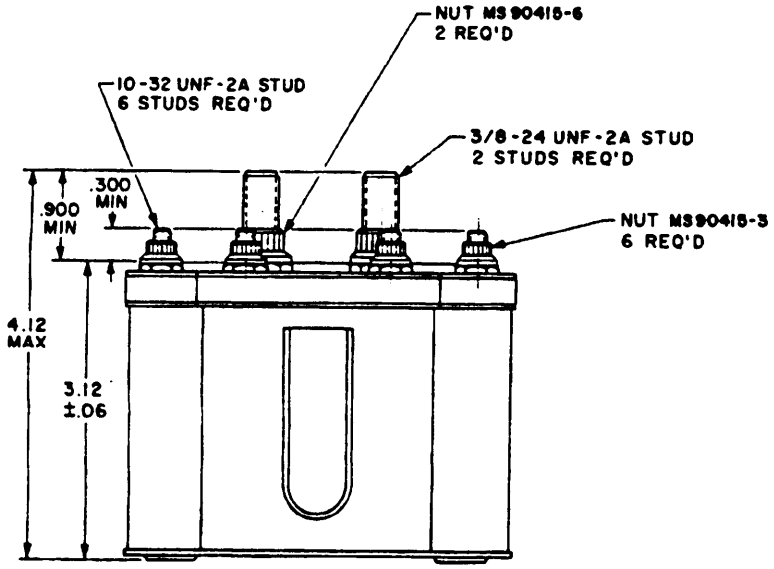
This specification has been approved by the Naval Air Systems Command, Department of the Navy. The complete requirements for the relay described herein shall consist of this document and the issue in effect of specification MIL-R-6106.




● MAIN CONTACTS
○ AUXILIARY CONTACTS

ALL STUDS CORROSION-RESISTANT STEEL, ROCKWELL C-20 MINIMUM HARDNESS




FSC 5945



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REQUIREMENTS:

1. The contractor shall have the following characteristics.

| | |
|---|---|
| Rated current (main contacts) | 400 amperes |
| Rated inrush current - 5 seconds | 1500 amperes |
| Rated current at 30 seconds | 1000 amperes |
| Maximum weight | 2.75 pounds |
| Rating auxiliary contacts | 5 amperes resistive |
| Auxiliary of contacts configuration | SPDT |
| Interrupting capacity | 4000 amperes |
| Economizing coil | 12-15 volts min |
| Maximum ambient temperature | 71°C |

2. The contractor shall include a standard Type A, B or H metal nameplate in accordance with MIL-P-15024, securely attached to the machine. The nameplate shall contain the following information. Abbreviations are permissible.

400 Ampere line contactor

| | |
|---|---|
| Rated voltage | 30 volts |
| Rated current (main contacts) | 400 amperes |
| Rated current (auxiliary contacts) | 5 amperes |
| Military part number | M6106/30-1 |
| Manufacturer's part number | |
| Manufacturer's name or trademark | |
| Serial no. | |
| Federal stock no. | |
| Date of manufacture | |
| Acceptance stamp | |

3. Where maximum dimensions are shown the part need not have the shape shown but shall be contained within the outline.

4. The relay shall be tested in a system consisting of:

   (a) 0-30 VDC power source

   (b) Resistive load bank that is adjustable from zero to 1,000 amperes.

NOTES:

1. Unless otherwise specified dimensions are in inches.

2. Tolerances are decimals .XX ± .03, .XXX ± .010.



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TESTS:

1. Qualification tests shall consist of only the tests listed in this specification sheet except for examination of product.

2. Examination of product shall be performed to the requirements of MIL-R-6106.

3. Closing voltage. The relay shall close when 12 VDC is applied to the coil terminals. Operation of the auxiliary contacts shall be monitored to insure correct operation.

(A) 4. Dropout voltage. The relay shall return to its de-energized position when the applied voltage is reduced to 6.5 volts.

(A) 5. Hold voltage. The relay shall be maintained in the "ON" position with a minimum applied voltage of 7.0 volts.

6. Potential drop. The drop in potential between the input and output terminals shall be measured at rated current. The potential drop shall be determined from an average of 5 readings of up to 2 minutes duration per reading. The drop for the average of 5 readings shall be less than .125 volt. No single reading shall sxceed .150 volt.

7. Potential drop after aging. After completion of test 9, the drop for the average of 5 readings shall be a maximum of .163 volt and no single reading shall exceed .244 volt.

8. Overload.

   (a) Main contacts. The relay shall be operated by a switch in series with the negative terminal and system ground. The current shall be 800 amperes for 500 cycles, consisting of 1 second "ON" and 10 seconds "OFF." After completion of this test, the relay shall be subjected to tests 3, 4 and 5.

   (b) Auxiliary contacts. The indicator lead shall be subjected to overload currents of 10, 20, 50, 75 and 100 amperes. A thermal circuit breaker type MS3320-5 shall provide circuit protection. After each application of overload current, it shall be determined that no damage has occurred to the relay.

   (c) Surge load. The relay shall be subjected to 500 operations with 2 minutes between surge loads and at a surge load and reduced voltage as shown in Figure 1. There shall be no welding of the contacts. After completion of this test, the relay shall be subjected to the potential drop test specified in test 5. The value of the potential drop shall not exceed the limits specified in test 5 by more than 10 percent. The relay shall also be subjected to tests 3 and 4.



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(d) Contact overload. The relay shall be subjected to 5000 cycles of overload contact current in accordance with the following table, and as specified in MIL-C-5026.

| LOAD | COIL VOLTAGE | TIME |
|---|---|---|
| 250 percent rated | 12 volts | 30 seconds |
| 200 percent rated | 28 volts | 4.5 minutes |

9. Interrupting capacity. The relay shall be so connected to the power source that a current of 4000 amperes is provided at the contact terminals. The contact shall make and break the interrupting current at 28 VDC (open curcuit) for a minimum of 10 cycles in accordance with the cycling time specified in test 7(a). A 1-ampere fuse shall be connected between the relay case and the system ground. Blowing of the fuse shall constitute failure.

10. Endurance.

(a) Sea-level. The relay shall be subjected to 5000 operations making and breaking rated current at rated voltage. The auxiliary circuits of the relay shall be connected to a 5-ampere resistive load. The test shall be conducted at a rate of 2 seconds "ON" and 4 seconds "OFF," or faster, so long as proportionate duty cycle is maintained. After completion of this test, the relay shall be subjected to the potential drop test specified in test 5. The value of the potential drop shall not exceed the limits specified in test 5 by more than 10 percent.

(b) Altitude. The preceding test shall be repeated at a pressure corresponding to an altitude of approximately 65,000 ft after completion of this test. The relay shall be subjected to the potential drop test specified in test 5. The value of the potential drop shall not exceed the limits specified in test 5 by more than 10 percent. The relay shall then be subjected to tests 3, 4 and 11.

11. Inductive load. The relay shall be subjected to 2000 operations of making and breaking rated current at rated voltage. The time constant of the load shall be 0.006 second. The first 1000 operations shall be at sea-level and the final 1000 operations at a pressure corresponding to an altitude of approximately 50,000 ft. The relay shall then be subjected to tests 3, 4, 6 and 11.

12. Dielectric strength. A potential of 1000 volts rms commercial frequency shall be applied between the terminals and other exposed metal parts for a period of 60 seconds. The coil ground connections, if any, shall be disconnected for this test. There shall be no breakdown.



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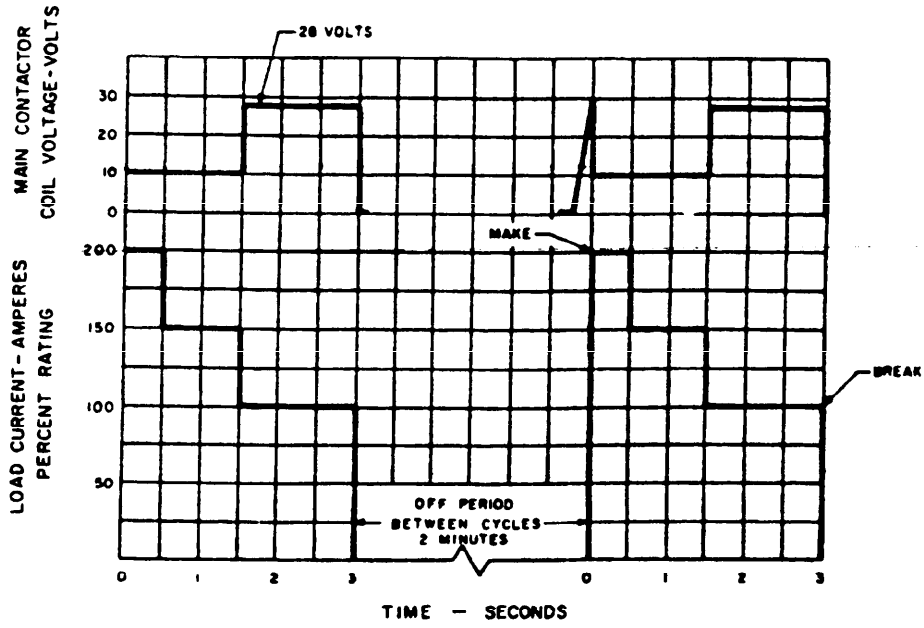



FIGURE 1.

13. Cold temperature. The relay shall be placed within a test chamber and cooled to and maintained at a temperature of $-54^{\circ} \pm 2^{\circ}C$ for a period of 72 hours. During the 72-hour period, the relay shall be cycled for a period of 30 minutes of each 24-hour interval. During the cycling period, the load current shall be 1/20 of the relay rating. The cycling period shall consist of 1 second "ON" and 5 seconds "OFF." At the end of this period and with the relay at the above specified temperature, the relay shall meet tests 3, 4 and 6. No failure or malfunctioning shall occur either during or as a result of the cycling procedure.

14. High temperature. The relay shall carry rated current at a coil voltage of 29 volts at maximum ambient temperature for a period of 3 hours. During the last hour of this period, the relay shall be subjected to the tests specified for tests 2, 3, 4 and 6. Following these tests, the relay shall meet the requirements specified for dielectric strength.

15. Shock. The relay shall be shock-tested in accordance with shock test, Procedure I, of MIL-STD-810. The relay shall be rigidly fastened, using the relay mounting means, to the drop carriage of the shock testing machine and subjected to single resilient impacts of 25g intensity of 6 to 9 milliseconds duration. Four shocks shall be applied in each direction along each of three mutually perpendicular body axes, two of each shall be with the main contactor coil de-energized and two with the coil energized. The relay shall be energized with 18 volts applied to the coil for 3 hours prior to and during the time of impact. Evidence of contact chatter shall be determined by an oscillograph. The total duration of contact chatter shall be less than 2 milliseconds.



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16. Acceleration. The relay shall be subjected to an acceleration force of 10g. The relay shall be mounted in the position most likely to cause malfunctioning and shall be accelerated for a minimum of 1 minute. During this test, the main contacts shall remain open with no voltage across the coil, and shall remain closed when the coil voltage is 18 volts. A test lamp or other indicating device shall be connected across the contacts while the relay is undergoing this test to determine the ability of the relay contacts to remain in the proper position.

17. Vibration. The relay shall be subjected to simple harmonic or circular motion applied to the mounting base. The maximum deflection shall be 0.06 to 0.07 inch. The frequency shall be varied from approximately 10 to 55 and back to 10 cycles per second, the period of variation being 1 to 5 minutes. The relay shall be mounted horizontally and vibrated continuously for 2 hours at 29 volts, full load: and then for 1 hour unenergized. The test shall be repeated with the axis revolved 90 degrees to the original axis. At least one of these tests shall be conducted with the plane of vibration parallel to the plane or planes of contact travel. Tests 3 and 4 shall be performed at the end of each 3-hour vibration period at C, 10, 20, 30, 40, 50 and 55 cycles per second, 0.06 - inch total excursion. The performance requirements specified in tests 3 and 4 shall be met.

18. Sand and dust. The relay shall be subjected to the sand and dust test, Procedure I, of MIL-STD-810. At the end of the test, the relay shall meet the requirements specified for tests 2, 3, 4 and 6.

19. Humidity. The relay shall be subjected to humidity tests, Procedure I, of MIL-STD-810. During the cycling procedure, a potential of 150 volts rms at commercial frequency shall be applied between the terminals and other exposed metal parts. The current leakage shall not be in excess of 0.1 ampere at any time during this test. After the relay has been removed from humidity, it should be dried for approximately 6 hours in a circulating air oven at approximately 57°C. Immediately after drying, the relay shall be subjected to and shall satisfactorily meet the test requirements for tests 2, 3, 4, 6 and 11.

20. Fungus resistance. The relay shall be subjected to fungus resistance test, Procedure I, of MIL-STD-810. No component part of the relay shall exhibit fungus growth. This test need not be conducted provided the manufacturer certifies that none of the material used on the relay will deteriorate or corrode under the fungus conditions of Procedure I.



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21. Salt spray. The relay shall be subjected to the salt spray test of MIL-STD-810, for 50 hours. After the relay has been removed from salt spray, it shall be dried for approximately 6 hours in a circulating air oven at approximately 57°C. Immediately after drying, the relay shall be subjected to and satisfactorily meet the test requirements for tests 2, 3, 4 and 6. No component part of the relay shall exhibit any signs of corrosion.

22. Explosion-proof (aeronautical). The relay shall be subjected to and shall satisfactorily meet the explosion-proof (aeronautical) tests, Procedure I, 3.1.1 (c), of MIL-STD-810. All operating tests shall be conducted at rated load with a time constant of 0.006. At the end of the test, the relay shall meet the requirements specified in tests 2, 3, 4 and 6.

Preparing Activity:
Navy - AS
(Project No. 5945-N499-1)



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